

Favorable effects of berry consumption on platelet function, blood pressure, and HDL cholesterol.



BACKGROUND: Berries are a particularly rich source of polyphenols. They also contain other bioactive substances, such as vitamin C. Previous studies indicated that the consumption of polyphenol-rich foods (eg, cocoa, tea, and red wine) may induce beneficial changes in pathways related to cardiovascular health. Whether the consumption of berries has similar effects is unknown. **OBJECTIVE:** We aimed to investigate the effects of berry consumption on hemostatic function, serum lipids, and blood pressure (BP). **DESIGN:** Middle-aged unmedicated subjects (n = 72) with cardiovascular risk factors consumed moderate amounts of berry or control products for 8 wk in a single-blind, randomized, placebo-controlled intervention trial. **RESULTS:** Berry consumption inhibited platelet function as measured with a platelet function analyzer (using collagen and ADP as platelet activator) [changes: 11% and -1.4% in the berry and control groups, respectively; $P = 0.018$, analysis of covariance (ANCOVA)]. Plasma biomarkers of platelet activation, coagulation, and fibrinolysis did not change during the intervention. Serum HDL-cholesterol concentrations increased significantly more ($P = 0.006$, ANCOVA) in the berry than in the control group (5.2% and 0.6%, respectively), but total cholesterol and triacylglycerol remained unchanged. Systolic BP decreased significantly ($P = 0.050$, ANCOVA); the decrease mostly occurred in subjects with high baseline BP (7.3 mm Hg in highest tertile; $P = 0.024$, ANCOVA). Polyphenol and vitamin C concentrations in plasma increased, whereas other nutritional biomarkers (ie, folate, tocopherols, sodium, and potassium) were unaffected. **CONCLUSION:** The consumption of moderate amounts of berries resulted in favorable changes in platelet function, HDL cholesterol, and BP. The results indicate that regular consumption of berries may play a role in the prevention of cardiovascular disease.